

Security+ 3.1 · Cloud Infrastructures

Master CyberSec — Official CompTIA Security+ (SY0-701) Comprehensive Study Guide · **Lesson 3.1 — Cloud Infrastructures (9:56)** · [Official YouTube Video ↗](#) · [Professor Messer Study Hub ↗](#)

Executive Lesson Summary: Cloud service models: IaaS (e.g., EC2) provides virtualized infrastructure where the customer manages the OS and above; PaaS (e.g., Heroku, AWS Beanstalk) offers a ready platform for application deployment; SaaS (e.g., Gmail) delivers fully managed applications. Serverless/FaaS (e.g., Lambda) runs code on demand without server management, billed per execution. Infrastructure as Code (IaC) tools like Terraform and CloudFormation enable version-controlled, auditable infrastructure. Security controls include IAM, security groups, and VPCs.

High-Yield Core Concepts Tested on Exam

- IaaS: customer manages OS and above; PaaS: only application; SaaS: fully managed by provider
- Serverless (FaaS): no server management, pay per execution
- IaC (Terraform, CloudFormation): infrastructure as versioned, auditable code
- IAM, security groups, and VPCs are key cloud security controls

Exam Tips & Classic Scenarios

- 'cloud model where user manages the OS and above' = IaaS
- 'runs code without managing servers' = serverless/FaaS
- 'infrastructure defined in code and versioned' = IaC

Typical Exam Scenario: A company rents virtual machines and manages the operating system themselves. This is an example of IaaS.

Beyond the Video: Official CompTIA Exam Concepts, Protocols & Traps (Lesson 3.1 — Cloud Infrastructures)

Why is this advanced coverage required for SY0-701? Video overviews provide foundational familiarity, but CompTIA tests practical engineering judgment and requires deep familiarity with the following protocols, acronyms, and operational nuances in this domain:

- **SAAS:** Software as a Service: Cloud delivery model where the provider hosts and manages the entire application and infrastructure, with customers consuming the service over the web.
- **PAAS:** Platform as a Service: Cloud model where the provider manages hardware and OS, providing a managed platform for application development and execution.
- **IAAS:** Infrastructure as a Service: Cloud computing model offering raw virtualized computing resources (VMs, storage, networking) managed at the OS level by the customer.

Classic CompTIA Traps & Question Framing:

- Avoid rote memorization: CompTIA exam prompts regularly adjust technical constraints requiring you to defend why one valid security control is the *MOST appropriate* or *FIRST step*.
- Scrutinize question qualifiers: 'without additional licensing costs' favors native OS tools; 'granular administrative command accounting' favors TACACS+ over RADIUS; 'non-repudiation' mandates asymmetric private key signing.

Practice Questions with In-Depth Solutions (10 Questions)

Detailed answer key with step-by-step technical rationale and comprehensive distractor breakdown provided below. Attempt answering independently first.

1. You rent only the virtual machine and manage the OS and everything above. What model is this?

- A)** SaaS
- B)** PaaS
- C)** IaaS
- D)** On-premises

2. Gmail, Office 365, and Salesforce are examples of which model?

- A) IaaS**
- B) PaaS**
- C) SaaS**
- D) FaaS**

3. Which cloud service model offers the greatest control over the infrastructure, including the operating system and middleware?

- A) SaaS**
- B) PaaS**
- C) IaaS**
- D) FaaS**

4. Which AWS service allows you to run code without provisioning servers?

- A) EC2**
- B) ECS**
- C) Lambda**
- D) Elastic Beanstalk**

5. In the SHARED RESPONSIBILITY model, who handles the PHYSICAL security of the datacenter?

- A) The customer**
- B) The cloud provider**
- C) Both equally**
- D) An external auditor**

6. Corporate scenario (Cloud Infrastructures): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.1), an analyst evaluates the following technical requirement: 'IaaS: you manage the OS. PaaS: only your application. SaaS: everything by the provider'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: IaaS: you manage the OS. PaaS: only your application. SaaS: everything by the provider.
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

7. Corporate scenario (Cloud Infrastructures): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.1), an analyst evaluates the following technical requirement: 'Serverless (Lambda): no server — pay per execution'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Implement the corresponding solution: Serverless (Lambda): no server — pay per execution...
- D)** Allow shared access among administrators with generic credentials

8. Corporate scenario (Cloud Infrastructures): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.1), an analyst evaluates the following technical requirement: 'IaC (Terraform): infrastructure as code — versioned, auditable'. What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: IaC (Terraform): infrastructure as code — versioned, auditable...
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

9. Corporate scenario (Cloud Infrastructures): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.1), an analyst evaluates the following technical requirement: 'cloud model where user manages the OS and above' = IaaS. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: 'cloud model where user manages the OS and above' = IaaS...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

10. Corporate scenario (Cloud Infrastructures): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.1), an analyst evaluates the following technical requirement: 'runs code without managing servers' = serverless/FaaS. What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: 'runs code without managing servers' = serverless/FaaS...
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

COMPREHENSIVE ANSWER KEY & DISTRACTOR BREAKDOWN

1) C — IaaS.

✓ **Why it's correct:** IaaS (Infrastructure as a Service) is the cloud computing model in which the provider offers virtualized infrastructure resources, such as virtual machines, storage, networks, and firewalls, while the customer manages the operating system, middleware, runtime, applications, and data. In the statement, the customer rents only the virtual machine and manages the OS and everything above, which exactly corresponds to the IaaS responsibility division as defined by NIST SP 800-145 and the CompTIA Security+ syllabus (domain 3.1 – Security Architecture and Design).

✗ **Why the others are wrong:**

- *SaaS:* SaaS (Software as a Service) is the model in which the provider delivers complete ready-to-use applications, managing the entire infrastructure stack, operating system, middleware, and the application itself. The customer only accesses the software via web interface or API, without managing the OS or any lower layer. In the described scenario, the customer manages the OS and everything above, which is incompatible with SaaS, where that responsibility belongs to the provider.
- *PaaS:* PaaS (Platform as a Service) offers a managed development and execution environment, including operating system, runtime, middleware, and database services. The customer manages only the applications and data, not the OS. In the statement, the customer manages the OS and everything above, which exceeds the customer's scope of responsibility in PaaS. Therefore, it is not the correct model.
- *On-premises:* On-premises (or on-prem) refers to IT infrastructure installed and maintained physically on the organization's own premises, without cloud involvement. The customer manages everything from hardware to applications, and there is no rental of a virtual machine from an external provider. The statement explicitly describes renting a virtual machine, which characterizes a cloud model, not on-premises.

2) C — SaaS.

✓ **Why it's correct:** SaaS (Software as a Service) is the cloud computing model where the provider delivers complete, ready-to-use applications, typically accessed via a browser or API, with all infrastructure, operating system, middleware, and application managed by the vendor. Gmail, Office 365, and Salesforce are classic examples of SaaS because users simply use the software without worrying about servers, updates, or maintenance. The question asks for examples of this model, and all three fit perfectly within the NIST definition and the CompTIA Security+ syllabus.

✗ **Why the others are wrong:**

- *IaaS:* IaaS (Infrastructure as a Service) provides virtualized infrastructure resources such as virtual machines, storage, and networks, on which the customer installs and manages their own operating system and applications. Gmail, Office 365, and Salesforce are not raw infrastructure; they are complete applications delivered as a service. Therefore, they do not fit IaaS, where the customer would have control over the OS and lower layers.
- *PaaS:* PaaS (Platform as a Service) offers a managed platform for developing and running applications, including OS, runtime, and middleware, but the customer is still responsible for developing or deploying their own applications. Gmail, Office 365, and Salesforce are final software ready for the user, not platforms for development. Thus, they are not examples of PaaS, but rather SaaS.
- *FaaS:* FaaS (Function as a Service) is a serverless computing model where the customer deploys individual functions that run on demand in response to events, without managing servers or containers. Examples include AWS Lambda, Azure Functions, and Google Cloud Functions. Gmail,

Office 365, and Salesforce are not serverless functions, but complete SaaS applications. Therefore, FaaS is not the correct model for the examples cited.

3) C — IaaS.

✅ **Why it is correct:** The correct answer is IaaS (Infrastructure as a Service). In the IaaS model, the cloud provider delivers fundamental computing resources — virtual machines, block storage, virtual networks, and firewalls — but the customer retains control over the stack from the guest operating system upward. This means the customer is responsible for installing and patching the OS, configuring middleware (application servers, databases, runtimes), managing identities and access policies, and protecting the data. This is the model that offers the greatest degree of control over the infrastructure among the options presented, because the customer can choose the Linux distribution, kernel version, web server, JVM, and so on. CompTIA classifies IaaS as the cloud service model that provides control over the underlying infrastructure, including the OS and middleware, while the provider manages virtualization, physical servers, storage, and the physical network.

❌ **Why the others are wrong:**

- *SaaS*: SaaS (Software as a Service) is the model in which the provider delivers a complete, ready-to-use application, such as Microsoft 365, Salesforce, or Google Workspace. The customer manages nothing in the stack: no access to the operating system, middleware, runtime, database, or infrastructure. All operational and security responsibility from the application down to the physical layer belongs to the provider. Therefore, SaaS offers the least control over the infrastructure, not the most. It is the opposite of what the question asks.
- *PaaS*: PaaS (Platform as a Service) provides a managed environment for developing and running applications, including the operating system, runtime, middleware, and database services already configured and maintained by the provider. The customer controls only the application and, in some cases, limited environment settings. Although it offers more control than SaaS, the customer does not manage the operating system or the middleware — exactly the elements cited in the question. Thus, PaaS does not meet the requirement of greatest control over the OS and middleware.
- *FaaS*: FaaS (Function as a Service) is a serverless computing model in which the customer writes individual functions and the provider fully manages the infrastructure, operating system, runtime, and scaling. The customer's control is limited to the function code and some trigger and memory settings. There is no access to the operating system or middleware, which makes FaaS the model with the least level of control over the infrastructure, not the most.

4) C — Lambda.

✅ **Why it is correct:** The correct answer is Lambda. AWS Lambda is Amazon's serverless compute service, where the customer provides only the code (function) and defines a trigger (event). AWS automatically provisions, scales, and manages all underlying infrastructure — servers, operating system, runtime, and capacity — without the user needing to create or manage instances. The billing model is per invocation and execution time, not per running server. Therefore, Lambda is the exact answer for 'run code without provisioning servers,' as per the serverless/FaaS concept tested in the CompTIA Security+ syllabus.

❌ **Why the others are wrong:**

- *EC2*: EC2 (Elastic Compute Cloud) is AWS's virtual machine service. The customer provisions instances, chooses the type, AMI, operating system, configures network, storage, and is responsible for patches and guest OS management. It requires provisioning and administration of servers (even if virtual), so it does not meet the requirement of running code without provisioning servers.

- *ECS*: ECS (Elastic Container Service) is a container orchestration service. Although it can be used with Fargate mode (serverless), ECS itself requires the customer to define clusters, tasks, services, and in EC2 mode, manage the underlying instances. It is not a purely serverless service by definition; the customer still needs to provision and manage compute capacity (either via EC2 instances or Fargate tasks, which require CPU/memory configuration). Therefore, it is not the most direct answer for 'run code without provisioning servers.'
- *Elastic Beanstalk*: Elastic Beanstalk is a PaaS platform that automates application provisioning and deployment, but under the hood it creates and manages resources such as EC2 instances, load balancers, and auto scaling groups. The customer still needs to choose instance type, environment, and configurations, and the application runs on provisioned servers (though managed). It is not a pure serverless model; there is server provisioning, albeit automated.

5) B — The cloud provider.

✓ **Why it is correct:** In the cloud computing shared responsibility model, the physical security of the datacenter — including facility access control, video surveillance, redundant power, climate control, fire prevention and suppression, and protection against natural disasters — is the exclusive responsibility of the cloud provider. CompTIA Security+ SY0-701 emphasizes that the provider is always responsible for 'security OF the cloud' (the underlying infrastructure), while the customer is responsible for 'security IN the cloud' (their data, configurations, identities, and applications). The customer has no physical access to the datacenter and therefore cannot implement physical controls; the responsibility is non-transferable and not shared in this specific aspect.

✗ **Why the others are wrong:**

- *The customer*: The customer is responsible for security IN the cloud, i.e., their data, access configurations, encryption of their information, identity management, and patching of their instances. However, they do not have physical access to the provider's datacenter and cannot install locks, cameras, or fire suppression systems. Assigning physical security to the customer would violate the shared responsibility model, as they do not control the physical environment. The question specifically asks about the PHYSICAL security of the datacenter, which is the provider's exclusive domain.
- *Both equally*: Shared responsibility is not divided equally in all aspects. It is segmented by layers: the provider handles physical security, virtualization infrastructure, and the underlying network; the customer handles the security of their data, applications, and identities. The physical security of the datacenter is a layer the customer cannot operate or directly audit, so there is no equal sharing. Saying both are equally responsible for physical security is technically incorrect and contradicts the model defined by CompTIA and major cloud providers (AWS, Azure, GCP).
- *An external auditor*: An external auditor performs independent compliance assessments, such as SOC 2, ISO 27001, or PCI DSS certifications, but does not execute the physical security of the datacenter. They verify whether the controls implemented by the provider are adequate, but they are not responsible for operating them. Auditing is a verification function, not an operational one. Therefore, assigning physical security to an external auditor confuses the oversight role with the execution role, which is a serious conceptual error in the context of the shared responsibility model.

6) B — Implement the corresponding solution: IaaS: you manage the OS. PaaS: only your application. SaaS: everything by the provider..

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.1 (Cloud Infrastructures): IaaS: you manage the OS. PaaS: only your application. SaaS: everything by the provider. This reflects the shared responsibility model: in IaaS the customer secures the OS and above; in PaaS the provider manages the OS and runtime while the customer secures the application; in SaaS the provider manages nearly the entire stack.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to incident detection and it breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials*: Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

7) C — Implement the corresponding solution: Serverless (Lambda): no server — pay per execution....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.1 (Cloud Infrastructures): Serverless (Lambda): no server — pay per execution. Serverless computing abstracts the underlying infrastructure, allowing developers to run code without provisioning or managing servers. The cloud provider handles scaling and patching, and billing is based on actual execution time and resources consumed.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls that rely solely on user behavior and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials*: Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

8) A — Implement the corresponding solution: IaC (Terraform): infrastructure as code — versioned, auditable....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.1 (Cloud Infrastructures): IaC (Terraform): infrastructure as code — versioned, auditable.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls, rely solely on user behavior, and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators with generic credentials*: Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

9) B — Implement the corresponding solution: 'cloud model where user manages the OS and above' = IaaS....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.1 (Cloud Infrastructures): 'cloud model where user manages the OS and above' = IaaS. In

IaaS, the cloud provider manages the physical infrastructure, networking, and virtualization, while the customer is responsible for the operating system, middleware, runtime, applications, and data. This shared responsibility model is fundamental to cloud security.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

10) A — Implement the corresponding solution: 'runs code without managing servers' = serverless/FaaS....

✓ Why it's correct: According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.1 (Cloud Infrastructures): 'runs code without managing servers' = serverless/FaaS.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.